

Stepper motor

DO

OUTPUT

MOVEMENT

What does it do?

A stepper motor turns in small, repeatable steps instead of spinning freely.

Energising its coils in sequence moves the shaft one step at a time and holds its position.

The ULN2003 board supplies the coil current for the common 28BYJ-48 motor shown here.

Uses



Pointer dial

DETECT

Read a changing value

DECIDE

Find its dial position

DO

Step the pointer there



Snack lift

DETECT

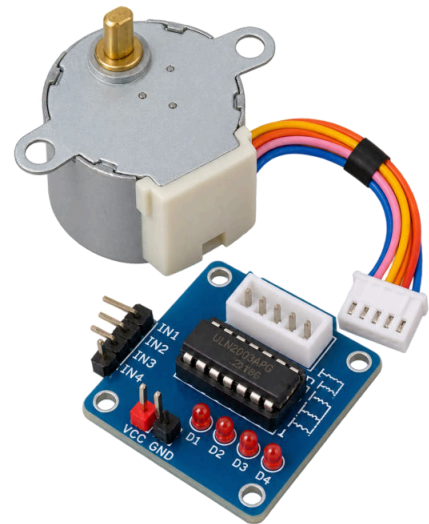
Snack request arrives

DECIDE

Which floor needs it

DO

Step the lift into place



Output shaft

Moves in measured steps

Five-wire plug

Connects to the driver

IN1 to IN4

Coil sequence

5V and GND

Driver power

Code

● Raspberry Pi

```
from gpiozero import OutputDevice
from time import sleep

coils = [OutputDevice(pin) for pin in (22, 27, 17, 4)]
sequence = ((1, 0, 0, 0), (0, 1, 0, 0),
            (0, 0, 1, 0), (0, 0, 0, 1))

for pattern in sequence:
    for coil, value in zip(coils, pattern):
        coil.value = value
    sleep(0.002)
```

● Raspberry Pi Pico

```
from picozero import Stepper

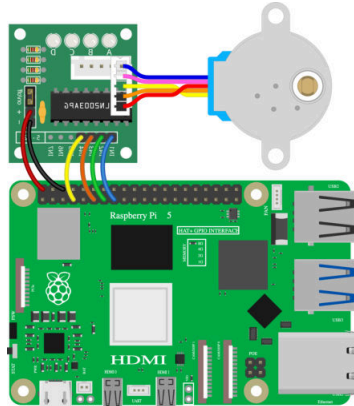
motor = Stepper((16, 17, 18, 19))

motor.step(100)
```

Repeat the Pi sequence to keep moving.
Reverse either board's sequence to turn the other way.

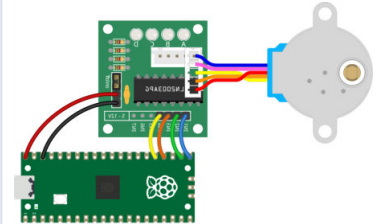
Wiring

RASPBERRY PI 5



IN1-IN4 → GPIO **22, 27, 17, 4**; driver →
5V and GND

PICO



IN1-IN4 → GP **16, 17, 18, 19**; driver →
VBUS and GND